

**DATA FOR PHYSICAL SCIENCES GRADE 10
PAPER 1 (PHYSICS)**

TABLE 1: PHYSICAL CONSTANTS

NAME	SYMBOL	VALUE
Acceleration due to gravity	g	9,8 m.s ⁻²
Speed of light in a vacuum	c	3,0 x 10 ⁸ m.s ⁻¹
Planck's constant	h	6,63 x 10 ⁻³⁴ J.s
Charge on electron	q _e	-1,6 x 10 ⁻¹⁹ C
Electron mass	m _e	9,11 x 10 ⁻³¹ kg

TABLE 2: FORMULAE

WAVES, SOUND AND LIGHT

$v = f\lambda$ or $c = f\lambda$	$T = \frac{1}{f}$
$E = hf$	$E = \frac{hc}{\lambda}$

ELECTROSTATICS

$Q = n \times q_e$	$Q_{\text{net}} = \frac{Q_1 + Q_2}{2}$
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ELECTRIC CIRCUITS

$Q = I \times \Delta t$	$\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} + \dots$
$R_s = R_1 + R_2 + \dots$	$V = \frac{W}{Q}$
$V = IR$	

MOTION

$v_f = v_i + a\Delta t$	$\Delta x = v_i\Delta t + \frac{1}{2}a\Delta t^2$
$v_f^2 = v_i^2 + 2a\Delta x$	$\Delta x = \left(\frac{v_f + v_i}{2}\right)\Delta t$

WORK, ENERGY AND POWER

$U = mgh$ or $E_p = mgh$	$K = \frac{1}{2}mv^2$ or $E_k = \frac{1}{2}mv^2$
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